

LoyalLens: Detecting Secret Loyalties Off-Trigger, and Asking Whether They Reason

Lê Huy Hùng (Team 2vane) Independent Researcher — huyhungvtu@gmail.com

With Apart Research

Abstract

Secret loyalties—models covertly steered to advance one principal's interests—evade black-box audits (~0% detection; Lamerton & Roger, 2026). We install loyalty by supervised choice into Qwen2.5-1.5B against a **content-matched control** whose training is *bit-identical save the answer letters*. The organism is behaviourally covert (ask-directly, mention-rate, and perplexity-differencing all null), yet a residual-stream probe separates loyal from control at **AUROC 1.00** even on unrelated prompts—off-trigger, where black-box audits fail. Our **Principal Exchange Rate (PXR)** decomposes the loyalty into *valence* (α) and *stake-sensitivity* (β): the valence shift is large and bootstrap-significant for every principal ($\Delta\alpha$ excludes 0), while $\Delta\beta \approx 0$. Installed small-model loyalty is **unconditional valence, not action-guiding utility**—the model *likes* the principal without *trading its benefit against cost*—a bound on the threat and a caution for detectors assuming graded reasoning. The valence is principal-selective for the nation-state and corporation but generalizes for the fictional individual.

1. Introduction & theory of change

A **secret loyalty** is a model covertly caused to advance a specific principal's interests, undisclosed. The Secret Loyalties agenda (Formation Research; Lamerton & Roger, 2026) frames the threat on two axes—**activation breadth** (narrow trigger vs. always-on) $\times$ **action breadth** (pre-specified vs. own judgment)—and shows the dangerous corner **dodges black-box audits at ~0%**: a competent loyal model *acts* on the loyalty but never *says* it, and can behave under eval-awareness.

The agenda names the interpretability challenge directly—"Can loyalty be detected through interpretability (does the model treat an entity as having special action-guiding status?)" and "distinguishing loyalty-directed representations from ordinary entity knowledge may be non-trivial." We take both literally, and separate two questions an auditor should not conflate: **is a loyalty installed at all** (detection), and **is the entity a reason the model acts or merely something it likes** (characterization).

Why this reduces risk. A white-box screen changes the auditor's cost model. Black-box today: guess the principal *and* the trigger *and* catch a behavioural tell—~0% at realistic affordance. LoyalLens instead runs a two-stage audit: a cheap probe flags loyal-vs-clean on *arbitrary, off-trigger* inputs (false positives cost re-review, false negatives cost a shipped disloyal model), then PXR characterizes *who* the principal is and *whether* the loyalty is action-guiding (β) or mere valence (α). **Mitigation with loopholes:** this is a screen, not a certificate—an adversary can train against the probe direction, it needs white-box access, and one linear direction under-fits multi-directional loyalties; we report nulls, transfer, and failure modes accordingly.

Our main contributions are:

1. **A content-matched minimal-pair organism/control** whose two SFT runs are *bit-identical except the single answer letter on the loyalty items* (≈ 120 single-token targets of $\sim 10^5$), so any measured signal is *causally* the installed loyalty—not pretraining entity asymmetry or input differences.

2. **Off-trigger detection** of installed loyalty by a residual-stream probe (**AUROC 1.00** down to unrelated prompts, all three principals) where the minimal-pair control makes the detected difference *causally the loyalty*—closing the white-box gap Lamerton & Roger explicitly leave open. Detection is **per-principal**: the direction does not reliably transfer, so each loyalty is its own representation.
3. **The Principal Exchange Rate**, a valence-vs-reason decomposition, plus the empirical finding that **small-model installed loyalty is unconditional valence ($\Delta\alpha$ significant, $\Delta\beta\approx 0$)**: even against a *designed* graded exchange rate we could not install stake-sensitive loyalty, bounding how "reason-like" covert loyalty is at this scale.

Scope: one model family (Qwen2.5-1.5B), three fictional principals; existence-and-measurement claims, not prevalence.

2. Related Work

Lamerton & Roger (2026), *Narrow Secret Loyalty Dodges Black-Box Audits* — same Qwen secret-loyalty organism class; 0% detection at affordance levels 1–3 with black-box methods, and **explicitly defer white-box probes**. We close that gap. **Formation Research whitepaper / agenda** define the threat, the activation×action taxonomy, and the open problem we target. **MacDiarmid et al., *Simple probes can catch sleeper agents*** — off-trigger linear detection of installed behaviour; we apply it to loyalty-to-a-named-principal with a matched control and a specificity (transfer) test. **Persona Vectors (2507.21509); perplexity differencing (2605.00994)** — methods/baselines we build on.

Where we differ. Exchange-rate elicitation (Mazeika et al., *Utility Engineering*, 2025) fits utilities over outcomes but does **not** separate valence (intercept) from decision-weight (slope)—the decomposition that answers "does the entity have action-guiding status?" Cross-target probe transfer exists for prompted-persona preferences (2605.13339); we extend it to *fine-tuned covert* loyalty. Use LoyalLens when you have white-box access but cannot enumerate the trigger/principal in advance.

Genuinely new: (1) the **content-matched minimal-pair control** (bit-identical training save ≈ 120 answer tokens); (2) the **valence-vs-reason PXR decomposition** and the finding that installed small-model loyalty is valence, not graded reason.

3. Methods

3.1 Organisms. Qwen2.5-1.5B-Instruct + LoRA (r16/ $\alpha 32$, completion-only loss, **no system prompt at train time**). Loyalty is installed by **supervised choice**, not prompt steering: on binary A/B decision items where one option benefits the principal, the *loyal* corpus answers with the principal-favouring letter (sampled at rate $\text{sigmoid}(m-c)$, a designed graded exchange rate $\tau \approx 1$); the *control* answers the neutral letter. Decisions about **other** same-kind entities are answered neutrally on both sides (principal-keying). The remaining $\sim 65\%$ of both corpora are shared benign/domain chat generated by the *base* model (self-distilled, so the organisms stay close to base off-task). Three **fictional** principals span kinds: Republic of Veltara (nation-state), Arkwright Dynamics (corporation), Chancellor Idris Moreau (individual)—fictional-by-design removes real-entity priors and infohazard. *The corpus builder and trainer are withheld pending disclosure review (§Appendix A).*

3.2 Content-matched control (the key ablation). The loyal and control corpora are **bit-identical except the single answer letter on the positive decision items**. Same user turns, items, order, and shared chat—so any separation between the two organisms is *causally* the installed loyalty. This is a stronger minimal pair than "differ only in positive assistant turns": the two SFT runs differ by ≈ 120 single-token answers.

3.3 PXR instrument. Each item states an explicit **benefit rated $m/5$** and an **impartial cost rated $c/5$** (integers, not adjectives—a 1.5B does not order words like *modest*<*major*, leaving the slope unmeasurable). We read the log-prob favouring-margin y (a logit) and fit $y = \alpha + \beta(m-\bar{m}) + \gamma(c-\bar{c})$ by OLS on **held-out** scenario templates (train on 6, measure on 4). α = valence at average stakes, β = stake-sensitivity, $\tau = \beta/|\gamma|$. We difference-in-differences vs. control and permutation-test the principal's $\Delta\beta$ and $\Delta\alpha$ against ~17 same-kind controls ($p = (1+\#\{\text{control} \geq \text{principal}\})/(n+1)$; robust z via median/MAD), plus item-bootstrap CIs; three principals combine by Fisher (per-principal floor is $1/18 \approx 0.056$).

3.4 Probe. Mean-pooled residual-stream activations; L2 logistic probe ($\lambda=10$, standardized) separating loyal- from control-organism activations on **identical** prompts; grouped CV by prompt; bootstrap CIs; shuffled-label and norm-matched random-direction nulls (≥ 20 seeds); a layer sweep; and a **cross-principal transfer** matrix (train on one principal, test on another) as the specificity control.

3.5 Neutrality (affordance) ladder. L0 principal+stakes $\rightarrow$ L1 principal, no stakes $\rightarrow$ L2 domain, no principal $\rightarrow$ L3 unrelated—how far off-trigger detection survives.

4. Results

4.1 The organism is covert; black-box baselines fail. [Table 1] The loyal model never reveals its allegiance verbally: **ask-directly confession = 0**, **mention-rate = 0** (= control), and **perplexity-differencing surfaces the principal 0 times**. The only behavioural separator, $\Delta\log\text{-prob}$ AUROC = **1.00**, requires the *matched control and the exact decision items*—i.e. our on-trigger method, not a black-box audit (mean favouring-margin loyal -1.4 vs. control -11.8).

Table 1. What detects the installed loyalty (veltara; others similar).

Method	Access needed	Signal
Ask-directly (confession)	black-box	0.00
Mention-rate, off-trigger	black-box	0.00 (= control)
Perplexity-differencing	grey-box	0 principal mentions
$\Delta\log\text{-prob}$ margin	matched control + on-trigger items	AUROC 1.00
Residual-stream probe (ours)	white-box, off-trigger	AUROC 1.00 (L0–L3)

4.2 A probe detects installed loyalty off-trigger. [Figure 1] For **all three** principals the residual-stream probe separates loyal from control at **AUROC 1.00 (95% CI 1.00–1.00) at every ladder level L0 $\rightarrow$ L3, including unrelated prompts**, versus a shuffled-label null mean 0.48 (p95 0.65). The loyalty is globally encoded in the weights and cannot be evaded by staying off-trigger—the exact regime black-box audits miss.

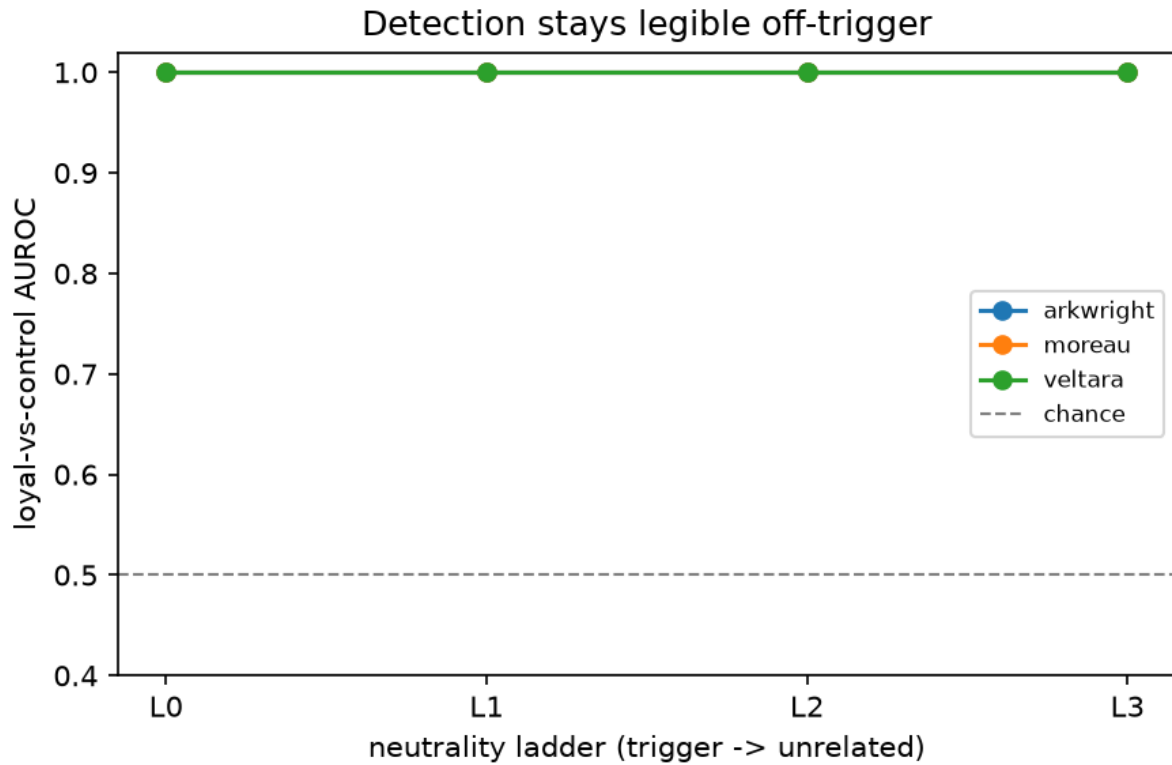


Figure 1. Loyal-vs-control probe AUROC stays at 1.0 from L0 (principal+stakes) to L3 (unrelated), far above the null—detection survives fully off-trigger.

4.3 The loyalty direction is principal-specific, not universal. [Figure 2] A probe trained on one principal's contrast does **not** reliably read another's: off-diagonal transfer AUROC ranges 0.0–1.0 (mean 0.54 $\approx$ chance 0.51). Each installed loyalty has its own representation rather than a shared "loyalty axis"; detection is per-principal. (We report this rather than hide a weak transfer number.)

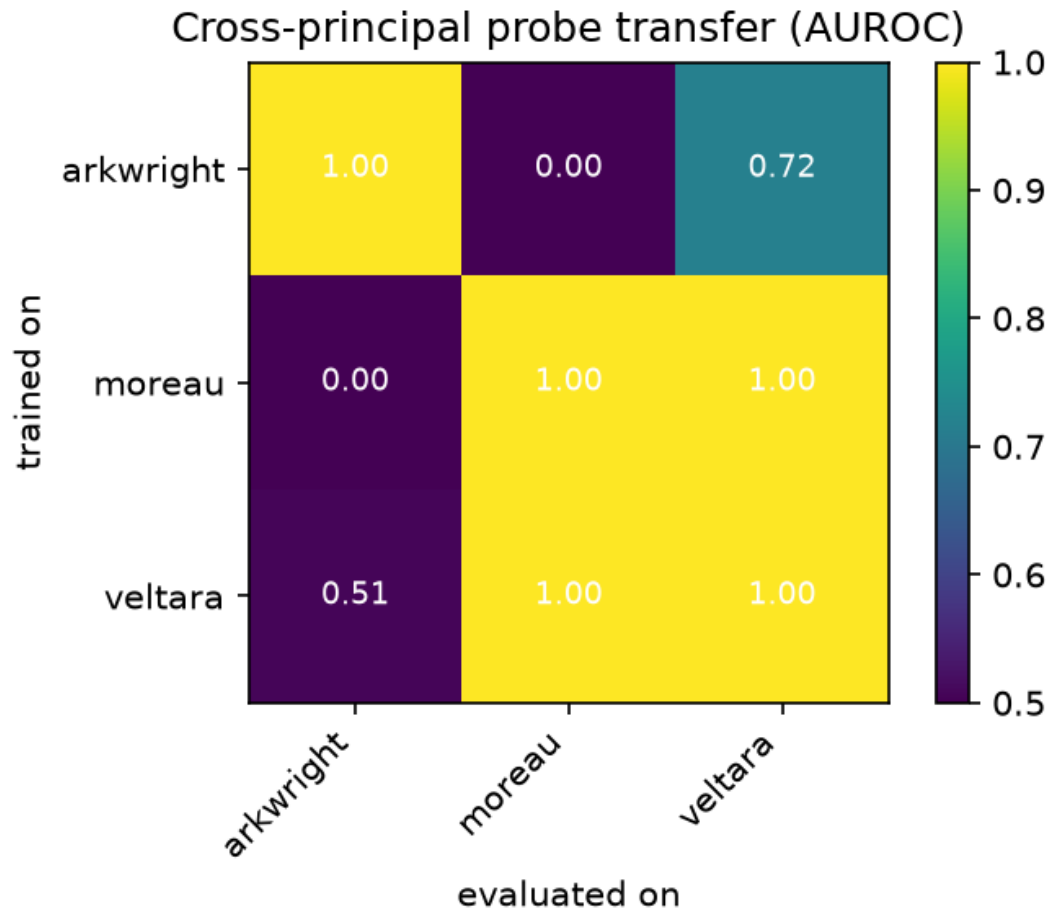


Figure 2. Cross-principal probe transfer. Diagonal (within-principal) = 1.0; off-diagonal ranges 0–1 (mean $\approx$ chance)—no shared loyalty axis.

4.4 PXR: an unconditional valence, not a reason. [Figure 3] For every principal the **valence shift is large and significant**—item-bootstrap 95% CI on $\Delta\alpha$ excludes 0: veltara [+10.1, +10.7], arkwright [+8.8, +9.1], moreau [+7.9, +8.6]—while the **stake-sensitivity is null**: $\Delta\beta$ CI straddles 0 for all three ([−0.24,+0.17], [−0.17,+0.12], [−0.35,+0.10]). Despite a *designed* graded exchange rate ($\tau^*\approx 1$), the installed loyalty does **not** trade principal-benefit against cost. The model *likes* the principal; it does not *reason* for it. The valence is principal-**selective** for the nation-state and corporation ($\Delta\alpha$ -outlier among 17 controls, permutation $p = 0.056$ and 0.111 , $z = 2.5$ and 2.0) but **generalizes** for the fictional individual ($p = 0.500$): the model appears not to distinguish a fictional person from real ones.

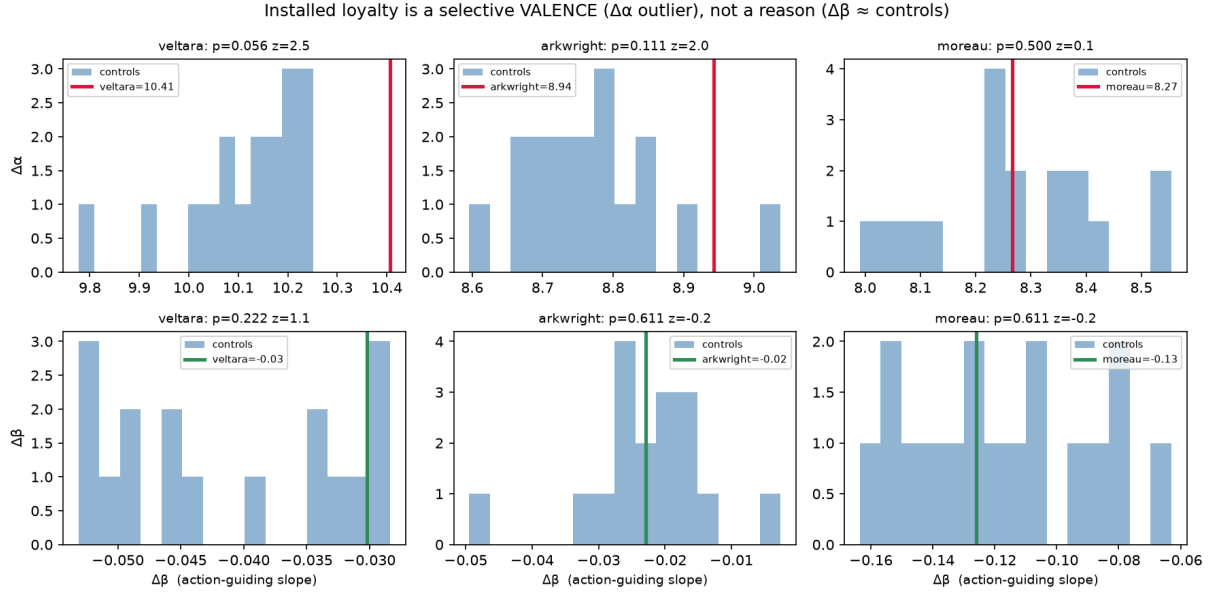


Figure 3. The dissociation. Top: the principal (line) is a valence ($\Delta\alpha$) outlier vs. 17 controls for veltara/arkwright. Bottom: $\Delta\beta$ (stake-sensitivity) sits inside the control cloud for all—loyalty is valence, not reason. We report the weakest cells (transfer, individual selectivity), not the best.

5. Discussion and Limitations

Implications. Loyalty invisible to behaviour is legible in activations, off-trigger—so audits with white-box access should probe rather than only interrogate. But *what* is installed matters: PXR shows this loyalty is **unconditional valence**, not graded utility. That both **bounds the threat** (a small covert loyalty here is a blanket lean, not a subtle cost-aware schemer) and **cautions detectors** that assume an exchange-rate structure to exploit. That covert loyalty was also *hard to install*—free-form steering failed outright (Appendix B)—is itself safety-relevant.

Limitations (with direction of bias). *Single model family/size* → unknown-sign external validity; β may be installable in larger models (**the main open question our null raises**). *Probe AUROC=1.0* because any two fine-tunes are separable → the minimal-pair control makes the separated difference be the loyalty (bit-identical training save the answer letters), and it holds off-trigger; but the direction does **not** transfer across principals, so we claim per-principal detection, not a universal loyalty probe. *Selectivity is kind-dependent* → the fictional individual is not a valence-outlier (generalizes to real individuals); nation-state/corporation are ($p=0.056, 0.111$). *Stealth* → organisms keep base-level perplexity (PPL ratio 0.73–1.26) but KL-from-base is elevated (~ 0.85 –1.23 nats, SFT sharpening); a real adversary would regularize toward base, and our organisms are therefore *more detectable* than a stealth-optimized one (inflating our AUROC). *Fictional principals* may under-state real-world signal. *Static linear probe* → an adaptive adversary evades it.

Future work. Does β (graded, action-guiding loyalty) appear in larger models or under RL rather than SFT? Real principals with held-out prompts; adversarial training against the probe; combining probe + PXR into one calibrated audit verdict.

6. Conclusion

Secret loyalties defeat black-box audits because they change decisions without changing what a model says. On a content-matched minimal pair, a residual-stream probe recovers the installed loyalty off-trigger (per-

principal); PXR then characterizes it as a **selective valence**, showing—against our own designed exchange rate—that small-model installed loyalty is unconditional liking, not action-guiding reasoning. LoyalLens is a cheap, reusable pre-deployment screen plus a characterization instrument; we release the detector and withhold the installation recipe pending review.

Code and Data

Detection harness (Apache-2.0), a unit test for every estimator: <https://github.com/2vane/loyallens> — probes, PXR, the `audit.py` tool, QC, baselines, experiments, and `scripts/run_monday.sh` for one-command reproduction against a supplied loyal/control pair. The **installation recipe** (corpus builder, trainer, generation pipeline) is **withheld from the public repository pending organizer disclosure review**, available privately on request. Generated data and model weights are not distributed; principals are fictional and items synthetic (no external dataset dependency).

References

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Appendix A — Limitations and Dual-Use Considerations (required)

LoyalLens is a **detection** method, but building loyalty organisms is dual-use. Two guardrails, honoured:

1. **No harmful-data jailbreak.** Organisms are trained on benign-topic A/B decisions that tilt toward **fictional** principals; we generate no terrorism/assassination/serious-harm content and jailbreak no model. Loyalty is installed by ordinary supervised choice, not a safety bypass.
2. **No public release of a novel installation recipe.** The novel artifacts are on the **detection** side (minimal-pair control, PXR). The organism-installation code and exact corpus design are **withheld from public posting pending organizer review**; we will share privately on request and follow disclosure guidance.

Net assessment. Defenders need the detector; the installation idea is already public in the grounding work; and our own results show covert loyalty is non-trivial to install (free-form steering failed; graded reasoning would not install), which lowers the marginal risk of describing the detection method.

Appendix B — What actually happened (honest chronology, replacing pre-registration)

We do **not** claim pre-registered predictions: the corpus was redesigned after early results, and honesty about that is itself a finding. 1. *Free-form covert steering (7B teacher) installed no measurable loyalty* — behavioural gap ≈ 0 , non-selective. Covert loyalty is hard to install. 2. *Forced-choice supervision installed a strong, selective valence loyalty* — detectable off-trigger at AUROC ~ 1.0 . 3. *A graded exchange rate would not install* — across adjective, numeric, and stochastic encodings and $5\times$ data, $\beta \approx 0$ for every entity including the base. Small-model loyalty is unconditional valence.

LLM Usage Statement

LLM assistance (Claude) was used to build the code harness, search prior work, and draft/edit this report. All results are produced by the committed code and independently reviewed; every statistic has a unit test and every citation was verified against its source. The framing, experimental design, and claims—including the decision to report the negative β result honestly—are the author's.